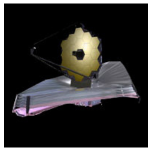

3201 - The Utility of Self-Consistent Models and Photochemistry in Understanding Transiting Planet Atmospheres

Cycle: 2, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

"Self-Consistent" or "Radiative-Convective Equilibrium" (RCE) models are the workhourse of planetary atmosphere modeling. Such models iterate to a solution for the mean atmospheric pressure-temperature profile and molecular abundances with all physics and chemistry known. Grids of RCE models are essential for predicting and explaining trends for transiting planet atmospheres, which retrievals cannot provide. These models are essential to the interpretation of exoplanet spectra, as it is only in comparison to RCE models that finding can be characterized as "expected" of "unexpected." Such models are the standard for data/model comparison for imaged planets and brown dwarfs, alongside retrievals. We propose to signficantly increase the usage and utility of RCE models for transiting planets by computing a grid to RCE model atmospheres in chemical equilibrium for every Cycle 1 and 2 target in the sub-Neptune to gas giant mass range. We will then push the science of RCE models forward by coupling two open source python tools, PICASO for atmospheric structure, and VULCAN for photochemistry, together. This coupling will yield the first open source tool, readily usable by any investigator, to examine the role of photochemisty and vertical mixing on atmospheric abundances, temperature structure, and spectra.

OBSERVING DESCRIPTION

Theory Proposal